THE CLOSURE OF A SUBMODULE IS THE INTERSECTION OF ITS TRANSLATES OF THE OPEN SUBMODULES

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Allowed standard prerequisites (editorial).

- The closure of a subset of a topological space as the intersection of the closed neighbourhoods, and openness of the translates x + M_lambda. Elementary topology of topological modules, as recalled in the chapter's section on linearly topologized modules. Justifies passing between the closure and the intersection of closed sets and the use of the translates of the open submodules as neighbourhoods of x.

Lemma 0.1. *Let R be a topological ring. Let M be a linearly topologized R-module and let M_λ, $\lambda \in \Lambda$ be a fundamental system of open submodules. Let $N \subset M$ be a submodule. The closure of N is $\bigcap_{\lambda\in\Lambda}(N + M_\lambda)$.*

Proof. Since each $N + M_\lambda$ is open, it is also closed. Hence the intersection is closed. If $x \in M$ is not in the closure of N, then $(x + M_\lambda) \cap N = 0$ for some λ. Hence $x \notin N + M_\lambda$. This proves the lemma. □

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